Computer Vision Project Proposal S'26

Team Name: Sharin-GAN

Project Title: Splatt3r for Human Body Resconstruction

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Introduction

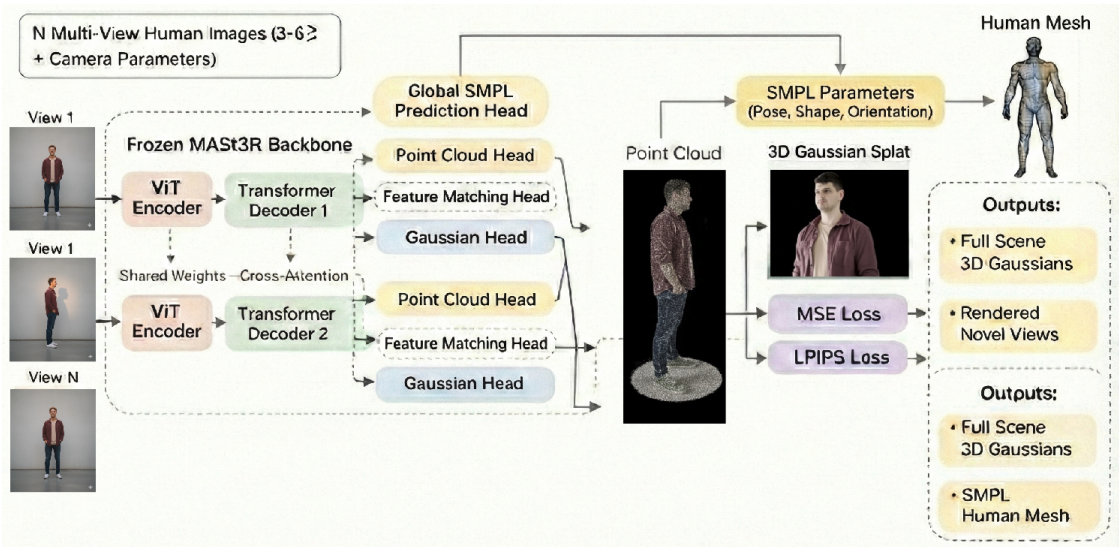
While Splatt3R focuses on object-centric 3D Gaussian splat reconstruction, many applications require reconstruction of deformable human bodies. Building on MASt3R's representations, an additional prediction head could be trained to output human-specific parameters such as SMPL body model coefficients or animatable Gaussian avatars. This project aims to adapt Splatt3R-like pipelines for human body reconstruction, leveraging multi-view consolidation via MASt3R-SfM and insights from CroCo-Man for human-centric priors.

Key Objectives:

1. Review methods for human body reconstruction from images, including CroCo-Man and Gaussian human avatars and add a new prediction head on top of MASt3R/Splatt3R to regress SMPL or avatar parameters.

2. Integrate with MASt3R-SfM to fuse N-view information into a single human reconstruction and evaluate on human-centric datasets.

Pipeline Overview



Input: Multi-view RGB images of humans.
Process:

1. Feature extraction using a frozen MASt3R backbone to obtain multi-view visual representations.

2. Multi-view feature consolidation using MASt3R-SfM to align features across views and estimate consistent 3D geometry.

3. Gaussian splat reconstruction following the Splatt3R methodology, where a dedicated Gaussian prediction head generates splat parameters (position, scale, rotation, opacity, and appearance features).

4. Addition of a human-specific prediction head operating on shared features:

   - Regression of SMPL body parameters including pose, shape, and global orientation.
   - Optional of animatable Gaussian avatar parameters.

5. Integration of human-centric priors inspired by CroCo-Man to improve reconstruction consistency and articulation handling.

Output: Structured human body representation (SMPL or Gaussian avatar) together with a photorealistic 3D Gaussian splatted reconstruction and an **optional animatable model**.

Evaluation Plan

Reconstruction Quality:

- **PSNR (Peak Signal-to-Noise Ratio):** Measures pixel-wise intensity similarity.
- **SSIM (Structural Similarity Index Measure):** Evaluates consistency in luminance, contrast, and structure aligned with human perception.
- **LPIPS (Learned Perceptual Image Patch Similarity):** Computes perceptual distance using features from a pre-trained deep network.

Human Body Accuracy: SMPL predictions will be evaluated using standard human pose metrics including **Mean Per Joint Position Error (MPJPE)** and, where available, per-vertex error (PVE) against ground-truth meshes or annotations.

Work Division

Member - 1 will focus on integrating the Splatt3R-style Gaussian reconstruction pipeline, including MASt3R-SfM processing, Gaussian prediction, rendering, and reconstruction evaluation.
Member - 2 will design and implement the human-specific SMPL prediction head, including parameter regression, loss design, and evaluation of structured human body reconstruction. (Although this is highly subjected to change :P)

Compute Resources

We have access to the ADA computing cluster equipped with four NVIDIA RTX 2080 Ti GPUs (12GB VRAM each). Pretrained MASt3R/Splatt3R models will be used to reduce training cost, with additional fine-tuning performed for the SMPL head and joint optimization. A total of 47-48 Gb VRAM is available with parallelization which we belive is good enough for this project. We tried to explore our available resources and roughly approximate the whole training head will take around 80M params which should be trainable with 24GB VRAM.

References

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[5] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. Black, "SMPL: A Skinned Multi-Person Linear Model," ACM Transactions on Graphics (SIGGRAPH Asia), 2015.

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